

MACHINE LEARNING



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TASK 1: Identify a scenario and perform a literature study

Scenario Analysis

Germany's equivalent of the UK's Pay-As-You-Earn(PAYE) system is the wage tax, known as 'Lohnsteuer'. It is a withholding mechanism through which income tax is collected from Germans' employment earnings at source. By each processed payroll, employers calculate, deduct and pay tax to the 'Finanzamt' (German tax office), similar to the Real Time Information(RTI) architecture of the coursework's dataset.

'Lohnsteuer' generated €248.9 billion in revenue in 2024, which makes it Germany's second-largest source of tax revenue after the 'VAT-system'. It also accounted for over 26% of total federal and state tax receipts (Destatis, 2025). Germany's wage tax system is one of the most data-rich in the world because it receives a lot of data continuously and therefore makes it good candidate for this paper.

The "Einkommenssteuergesetz(ESStG)" is the legal basis of the "Lohnsteuer." Germany has a modern income tax system. This includes a zero-rate band covering earnings, called 'Grundfreibetrag'(basic personal allowance), set at €11,784 in 2024. Whereby taxes increase from 14% to 42% for people who earn between €12,349 and €68,481. Additionally, the highest tax rate is defined at 45%.

The average single German worker paid 37.4% in taxes (OECD 2024). Whereby the average is predicted with only 25% accuracy. This places Germany among the countries with the most heavily taxed labour markets in the developed world. The methodological insights derived from German machine learning (ML) research can be directly applied to the mean-pay prediction task conducted herein, as both Germany's 'Lohnsteuer' and the UK's PAYE generate monthly payroll data across various industry sectors.

The "Steuerklassen" framework is a key part of how the German system works. It puts workers into one of six tax brackets, from I to VI. It depends on whether they are married and how many people in their household are working. However, it attracts significant criticism from economists and policymakers. The OECD(2022) demanded changes to Germany's joint taxation system, known as "Ehegattensplitting". The progressive rate structure gives single-earner households the highest tax break and makes it less likely that secondary earners, such as women, will work full-time or increase their hours.

The Ifo Institute also states that Germany is one of the few OECD countries whose tax system actively keeps gender inequality in the job market. The gender pay gap was 18% in 2023(Destatis, 2024). Additionally, Fuest, Peichl, and Siegloch's[1] study reveals that changes in corporate taxes in Germany disproportionately affect low-skilled, young, and female employees. A distributional pattern that is highly probable to exist in the industry-level pay data examined in this coursework. Therefore, the fact that Germany's tax system fails to address these structural problems is a political concern and a known source of bias in any machine learning model that uses German payroll data.

Another structural problem is Germany's trouble in accurately predicting "Lohnsteuer" revenues. Götttert and Lehmann(2021)[4] conducted a comprehensive examination of this matter. Their investigation into the inaccuracies in forecasts produced by the 'Arbeitskreis Steuerschätzungen'(government body responsible for official projections) indicated that these discrepancies are mainly due to incorrect calculations of the fundamental tax

base(particularly wages and employment rates) rather than to flawed elasticity assumptions. This is a key point in this coursework because it shows that the quality of the input dataset is much more important than how advanced the model is. The Mean Pay dataset has errors and missing values purposely to show how this kind of tax base data quality problem can happen. The pre-processing steps in Task 3 are therefore highly important. They directly replicate the data cleaning challenge that undermines real-world tax forecasting in Germany.

Germany's digital tax infrastructure, particularly the ELStAM('Elektronische LohnSteuerAbzugsMerkmale')-system, was introduced in 2013. It replaced the old paper-based 'Lohnsteuerkarte' system with a centralised electronic database of employees' tax information. At each payroll cycle, employers retrieve each worker's tax information, including tax class and allowances, from the federal system. While this creates a structured, machine-readable data stream, it also makes the system vulnerable to evasion and reporting inconsistencies. Lahann, Scheid, and Fettke(2019)[2] demonstrate that transactional payroll and accounting datasets are well-suited to machine learning(ML)-based compliance detection. They use decision tree classifiers to identify value-added tax(VAT) violations in German-standard accounting records. Their finding that tree-based models outperform simple statistical methods for categorical and mixed-type tax data is important to the coursework(i.e.,task 4).

Tax avoidance creates a discrepancy between expected and actual tax revenue. Delgado et al. (2023) show that artificial neural networks can find nonlinear links between tax avoidance and earnings management in European countries, that standard linear regression cannot. This highlights how important machine learning (ML) models are when working with payroll data and gives guidance for task 4.

A tax forecasting model that uses genetic algorithms and agent-based simulation is proposed by Alexi et al. (2024). They demonstrate that accounting for population heterogeneity is more effective than using aggregate approaches. This insight supports the use of clustering in Task 5, since grouping employees by sector can reveal latent structures obscured by aggregate data. Overall, these findings emphasise the importance of Tasks 2 and 3, highlighting that ML methods outperform traditional approaches when data quality is sufficient.

While all the reviewed studies demonstrate the effectiveness of machine learning in tax-related applications. Their methodological focus differs significantly. For example, tree-based models(Lahann et al., 2019) are interpretable and well-suited to structured payroll data, whereas neural networks(Delgado et al., 2023) can capture non-linear relationships, but lack transparency. Similarly, econometric approaches(Götttert & Lehmann, 2021) prioritise data quality over model complexity, whereas agent-based simulations(Alexi et al., 2024) model behavioural heterogeneity. This comparison illustrates the significant trade-off between interpretability, predictive power, and the amount of data required. Therefore, directly informs the modelling choices in this coursework.

Summarised papers in tables

Ref Number and Year of the paper	Fuest, C., Peichl, A. and Siegloch, S. (2018)
Country/Region	Germany
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Corporate income tax; wage incidence (Einkommensteuer / Lohnsteuer)
Machine Learning Algorithms Applied	Difference-in-differences (DiD), event study and panel regression.
What are the key findings or contributions of this paper	Workers account for around 50% of the corporate tax burden. Low-skilled, young and female employees take on an even larger share. The incidence effect is stronger in firms with collective bargaining agreements.
What insight did you gain out of the paper	It has been found that tax changes are not spread evenly through wages based on worker characteristics. This has highlighted the potential for structural bias in industry-level pay data in this coursework. It has been understood that different Lohnsteuer and PAYE responses will be shown by industries with different gender and skill compositions, and that these responses cannot be captured by a single regression line.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study only covers the local business tax (Gewerbesteuer), not the Lohnsteuer schedule. Additionally, the DiD method cannot easily be extended to predict future values, as required for Task 4. A supervised machine learning (ML) approach, trained on longer time-series payroll data, could complement these findings.

Table 1: Summarised findings of first paper [1], 2018

Ref Number and Year of the paper	Lahann, J., Scheid, M. and Fettke, P. (2019)
Country/Region	Germany / EU
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	VAT and payroll-related accounting data
Machine Learning Algorithms Applied	Decision trees, ML classification
What are the key findings or contributions of this paper	Decision tree classifiers can successfully identify VAT compliance violations in German accounting data. Tree-based rules can be understood by auditors and can be applied to new data.
What insight did you gain out of the paper	For the prediction task in this coursework, tree-based methods (including Random Forest and Gradient Boosting variants of decision trees) are strong candidates because the feature space includes continuous pay variables and categorical industry labels, which is a mixed structure that suits these algorithms.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study addresses binary detection (violation/no violation) and does not quantify the magnitude of the revenue lost through under-reported tax. Extending the study to a regression framework to estimate the scale of under-reported tax would substantially increase its operational usefulness for tax authorities, as this would be directly analogous to predicting mean pay values in Task 4.

Table 2: Summarised findings of second paper [2], 2019

Ref Number and Year of the paper	Delgado, F.J., Fernández-Rodríguez, E., García-Fernández, R., Landajo, M. and Martínez-Arias, A. (2023)
Country/Region	Germany, UK, France, Italy, Spain
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Corporate income tax / effective tax rate (ETR)
Machine Learning Algorithms Applied	Artificial neural networks (ANN), nonparametric regression
What are the key findings or contributions of this paper	ANN regressions reveal nonlinear relationships between earnings management and corporate tax avoidance. These results are consistent across all five major EU economies, including Germany. Standard linear models fail to detect these patterns.
What insight did you gain out of the paper	This paper justifies the use of artificial neural network (ANN)-style or nonlinear machine learning (ML) models over simple linear regression for Tasks 4 and 5 of this coursework. The principle that income and tax variables rarely conform to linear relationships is confirmed empirically at the EU scale.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study period ends in 2015. This excludes the impact of post-2017 EU anti-tax avoidance directives, ELStAM digital reporting reforms and the shock to corporate tax data caused by the onset of the pandemic. Updating the ANN models with a decade of additional data would reveal whether the identified nonlinear relationships remain stable through structural breaks.

Table 3: Summarised findings of third paper [3], 2023

Ref Number and Year of the paper	Götttert, M. and Lehmann, R. (2021)
Country/Region	Germany
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Income tax and Lohnsteuer revenue forecasting
Machine Learning Algorithms Applied	Econometric error decomposition, regression analysis of forecast components
What are the key findings or contributions of this paper	Errors in tax revenue forecasts by Germany's official estimation working group are primarily driven by mis-estimation of the tax base, rather than errors in elasticity. This finding challenges the conventional focus on refining model parameters at the expense of data quality.
What insight did you gain out of the paper	This paper is the most directly applicable to this coursework. It explains why the corruption and missing data in the COMP1886 dataset cause forecasting problems, and why pre-processing in Task 3 is essential; poor data will produce poor predictions regardless of the ML algorithm applied in Task 4.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The paper does not test whether machine learning (ML)-based forecasting could outperform the traditional econometric methods used by the official forecasting group. Given the richness of ELStAM data, however, this comparison would be both feasible and highly relevant to policy.

Table 4: Summarised findings of fourth paper [4], 2021

Ref Number and Year of the paper	Alexi, A., Lazebnik, T. and Shami, L. (2024)
Country/Region	Cross-national / stylised economies
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Income tax revenue forecasting
Machine Learning Algorithms Applied	Genetic algorithm (GA), heterogeneous agent modelling, ML-based optimisation
What are the key findings or contributions of this paper	A microfounded general equilibrium (GA) model with heterogeneous taxpayer populations produces more accurate tax revenue forecasts than aggregate econometric models. It has been shown that modelling sub-population variation is critical for forecast quality.
What insight did you gain out of the paper	Both Germany's Lohnsteuerklassen system and the industry-sector breakdown in this coursework dataset encode the kind of taxpayer heterogeneity that is critical to capture, as shown in [5]. This directly motivates the clustering approach in Task 5, whereby employees are grouped by behavioural and structural similarity before modelling to reveal patterns that a single undifferentiated model would miss.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The model has only been validated on stylised, simulated economies. Applying it to Germany's actual Lohnsteuer time series data, which is publicly available in aggregate form from Destatis, would provide crucial real-world validation of the heterogeneous agent approach and produce findings of immediate policy relevance.

Table 5: Summarised findings of fifth paper [5], 2024

References:

- [1] Fuest, C., Peichl, A. and Siegloch, S. (2018) 'Do higher corporate taxes reduce wages? Micro evidence from Germany', *American Economic Review*, 108(2), pp. 393–418. Available at: <https://www.aeaweb.org/articles?id=10.1257/aer.20130570> (Accessed: 24 March 2026).
- [2] Lahann, J., Scheid, M. and Fetteke, P. (2019) 'Utilizing machine learning techniques to reveal VAT compliance violations in accounting data', in *Proceedings of the 2019 IEEE 21st Conference on Business Informatics (CBI)*, Moscow, Russia, 9–11 July 2019. IEEE. Available at: <https://ieeexplore.ieee.org/abstract/document/8808015> (Accessed: 24 March 2026).
- [3] Delgado, F.J., Fernández-Rodríguez, E., García-Fernández, R., Landajo, M. and Martínez-Arias, A. (2023) 'Tax avoidance and earnings management: a neural network approach for the largest European economies', *Financial Innovation*, 9(1), pp. 1–25. Available at: <https://link.springer.com/article/10.1186/s40854-022-00424-8> (Accessed: 24 March 2026).
- [4] Göttert, M. and Lehmann, R. (2021) *Tax revenue forecast errors: wrong predictions of the tax base or the elasticity?* CESifo Working Paper No. 9148. Munich: CESifo. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3872387 (Accessed: 24 March 2026).
- [5] Alexi, A., Lazebnik, T. and Shami, L. (2024) 'Microfounded tax revenue forecast model with heterogeneous population and genetic algorithm approach', *Computational Economics*, 63(5), pp. 1705–1734. Available at: <https://link.springer.com/article/10.1007/s10614-023-10379-2> (Accessed: 24 March 2026).

TASK 2: Exploratory Data Analysis (EDA)

Hypothesis 1: Sector-Level Relationship between Employment and Mean Pay

This hypothesis examines whether industries with higher employment levels also have higher average pay. Therefore, both given datasets were combined at the sector level, and Pearson correlations were calculated across industries. The number of employees and average pay across sectors and over time were measured using the Pearson correlation coefficient (r) to assess linear relationships.

There was a strong positive correlation in fields such as health and social work ($r \approx 0.98$) and professional and scientific activities ($r \approx 0.92$). Positive correlations (e.g. health: $r \approx 0.98$, professional services: $r \approx 0.92$) suggests that wages rise alongside employment in high-skill sectors. In contrast, negative correlations in manufacturing ($r \approx -0.51$) and retail ($r \approx -0.56$) indicate the opposite.

The link between employment and pay differs across various sectors because of differences in productivity and skills. This means that a single global model could distort the data. However, these correlations may be affected by external factors, such as inflation or policy changes, and should not be taken as evidence of causation.

Hypothesis 2: Macro-level co-movement of total employment and average pay

This hypothesis explores whether an increase in total employment is associated with rising average pay overtime. The combined monthly statistics demonstrate a noticeable positive correlation ($r \approx 0.84$) between the two metrics, supported through an increasing trend. However, short-term fluctuations indicate an influence of external factors, suggesting that aggregate trends may mask variations within sectors.

Although, overall characteristics provide valuable information by combining those with industry-specific data is crucial for improving accuracy and avoiding aggregation bias. Although aggregate trends can be informative, they can also obscure sector-specific differences, highlighting the need for more detailed modelling.

Visualisations of EDA

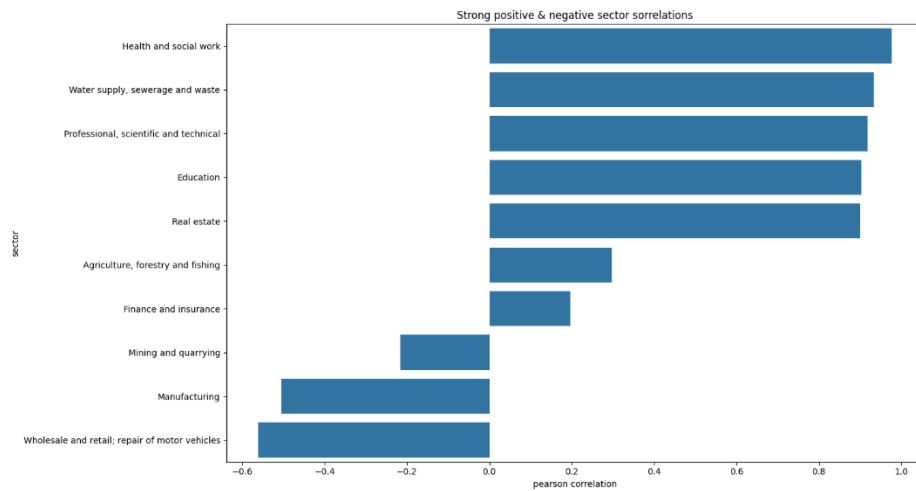


Figure 1: EDA Hypothesis 1 – horizontal bar chart Pearson correlation between PAYE employee count and mean pay

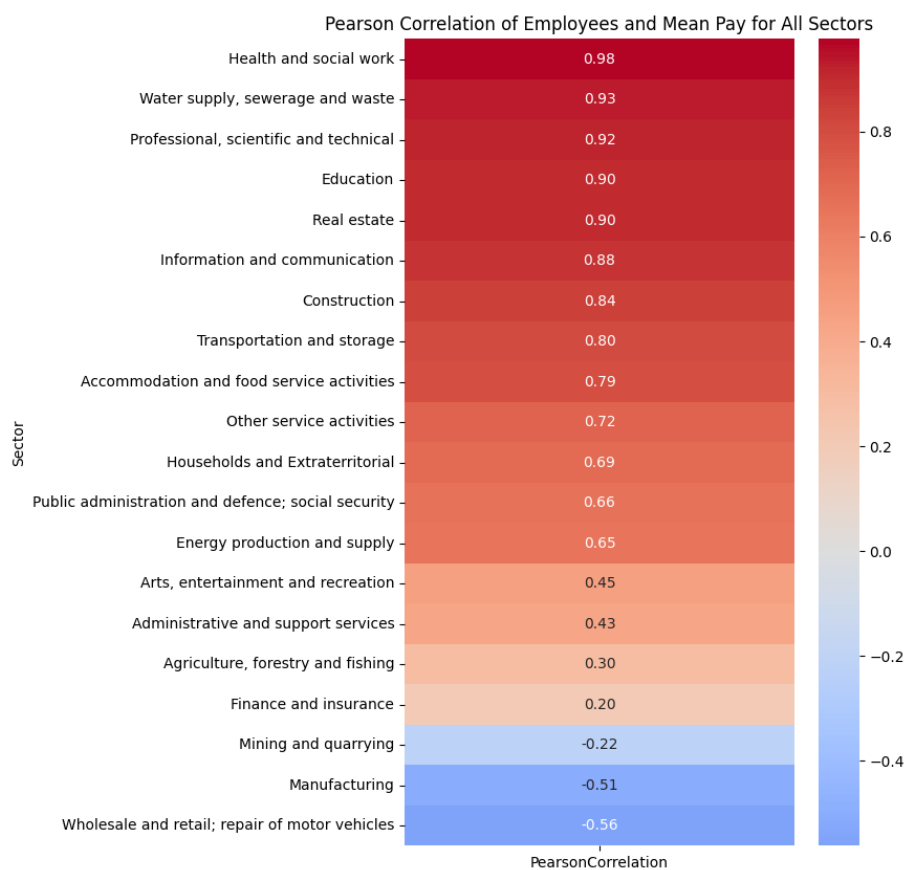


Figure 2: EDA Hypothesis 1: Heatmap Pearson correlation

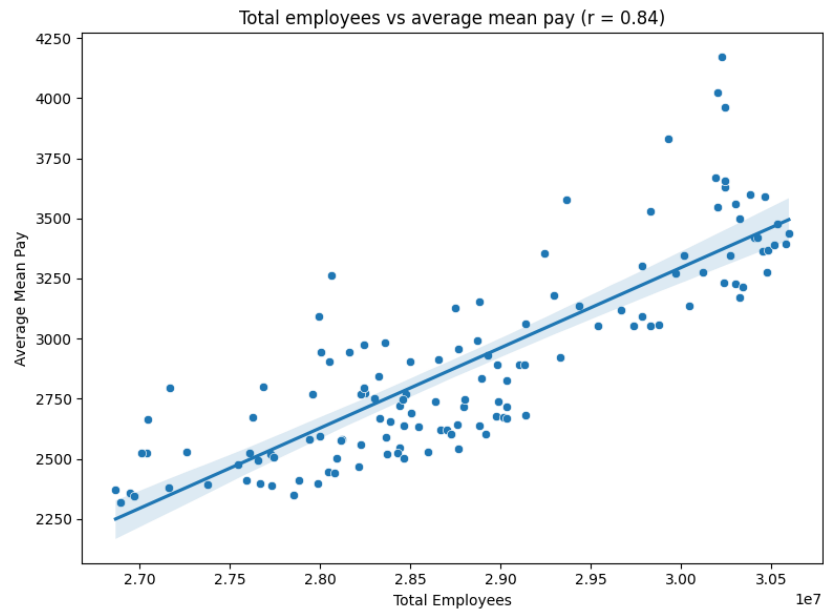


Figure 3: EDA Hypothesis 2 – scatter plot with OLS regression line

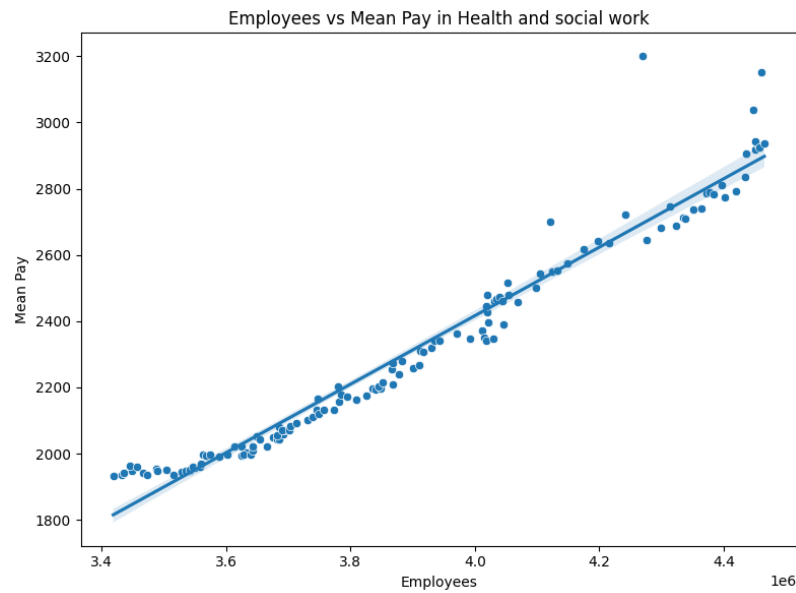


Figure 4: EDA Hypothesis 2 – scatter plot with OLS regression line 2

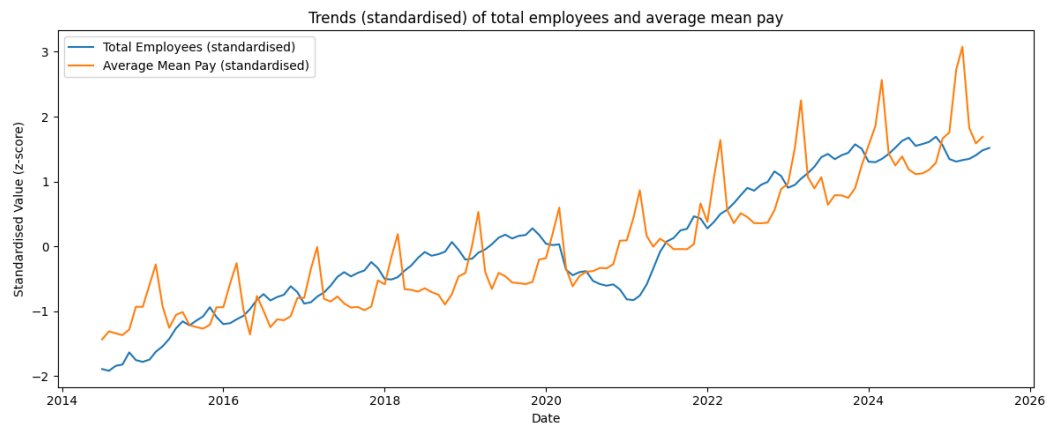


Figure 5: EDA Hypothesis 2 – dual standardised (z-score) time series of total PAYE employees and average pay

TASK 3: Data Pre-Processing

Technique 1: Missing Value Imputation

Missing values in the 'Mean_Pay' dataset were addressed by using linear interpolation. While leveraging the time-series structure and assuming gradual changes between months. This method was chosen because wage data usually changes gradually rather than showing sudden fluctuations.

Any remaining missing values, particularly at the boundaries of the dataset where interpolation was not feasible, were replaced using sector-specific mean imputation. This ensured that the data was complete while minimising distortion of the overall distribution. However, variability can be reduced through mean imputation, which introduces bias when missing values are not truly random.

Alternative methods i.e., of row deletion and forward filling was avoided due to data loss and bias. Whereby linear interpolation better preserves temporal continuity. Therefore, it makes it more suitable than forward-filling, that would artificially spread-out previous values and change short-term trends.

Technique 2: Data Transformation and Feature Preparation

This technique converted the column 'Date' to datetime format and set an index to enable a time-series analysis. Additionally, all sector columns were converted to numeric format. Invalid values have been replaced with NaN prior to imputation.

Both datasets were aligned by merging the common attributes. This ensures consistency between the input feature of employment and the target variable of mean pay. The StandardScaler was applied to normalise the features, setting the mean to 0 and the standard deviation to 1. This prevents scale dominance in distance-based algorithms such as K-Means.

These steps improve the quality of the data, keep the temporal structure and let machine learning models find meaningful patterns. This directly improves Task 4's predictive performance. StandardScaler transforms features to a mean of 0 and standard deviation of 1. This ensures that variables contribute equally to distance-based algorithms such as K-Means, which would otherwise be dominated by larger-scale features.

Visualisations & Findings of Data Pre-processing

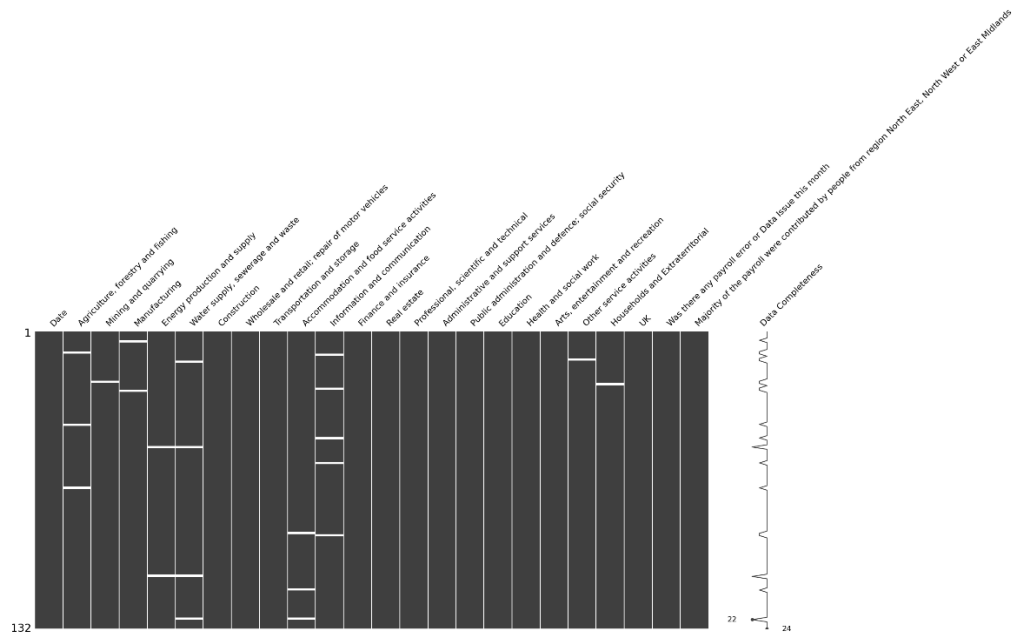


Figure 6: Missing Values in Dataset 'Mean Pay'

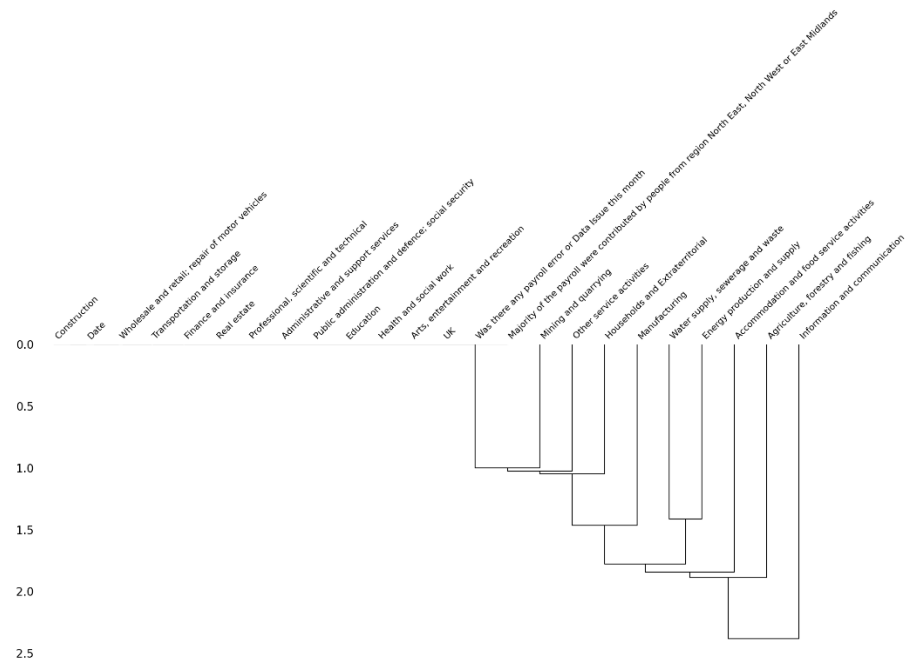


Figure 7: Missing Values Illustration 2

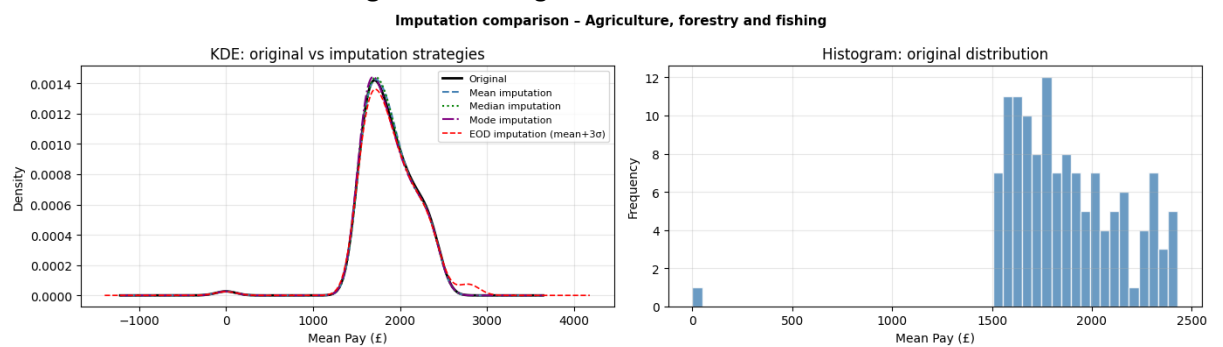


Figure 8: KDE comparison for sector: 'Agriculture, forestry and fishing' using the original, mean, median and mode strategies, as well as EOD imputation.

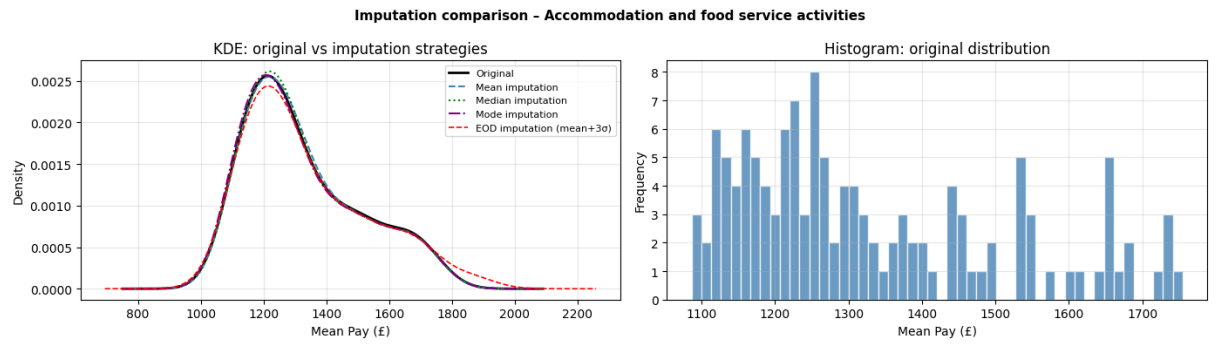


Figure 9: KDE comparison for sector: Accommodation and food service activities

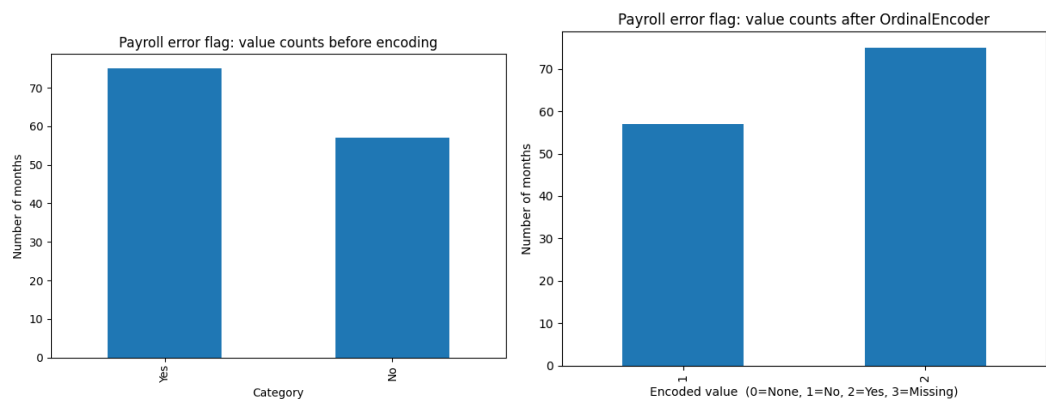


Figure 10: Payroll error flag value counts before and after OrdinalEncoder (1=No, 2=Yes).

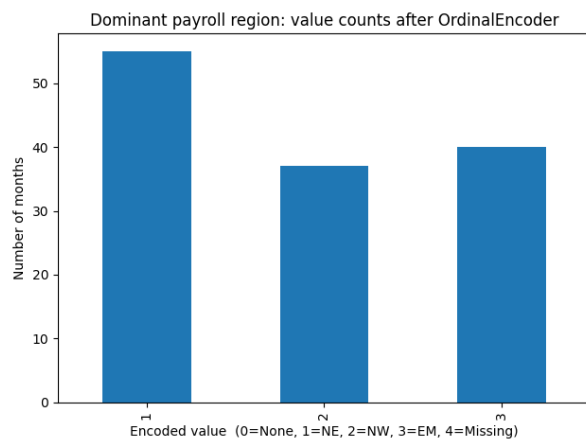


Figure 11: dominant payroll region

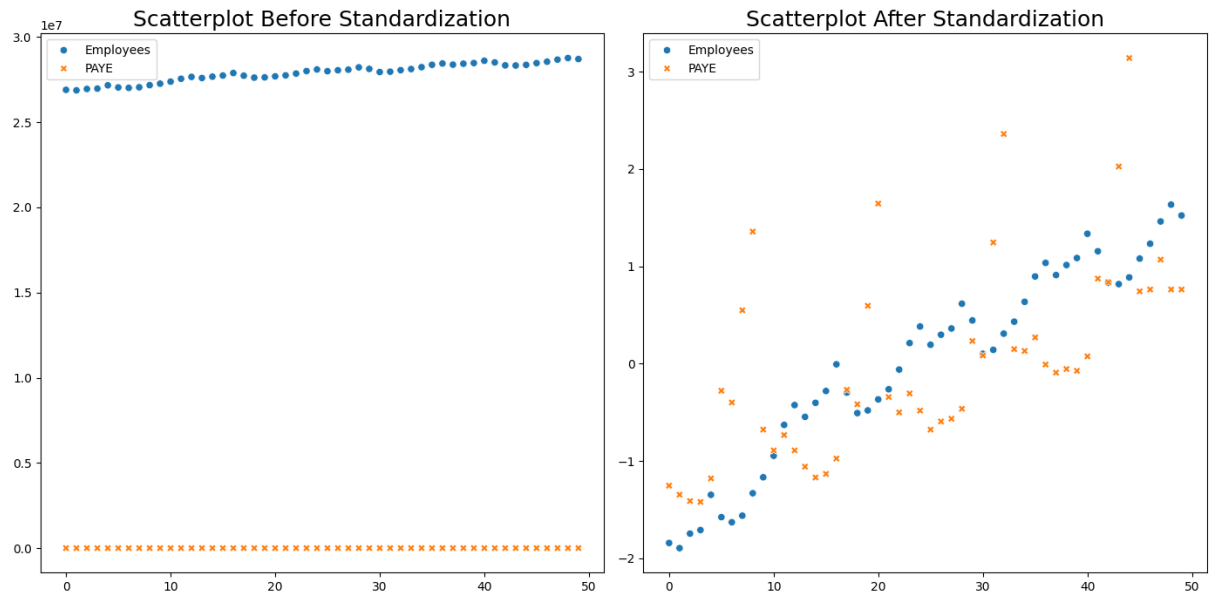


Figure 12: Scatterplots before and after StandardScaler applied to Employees and PAYE columns.

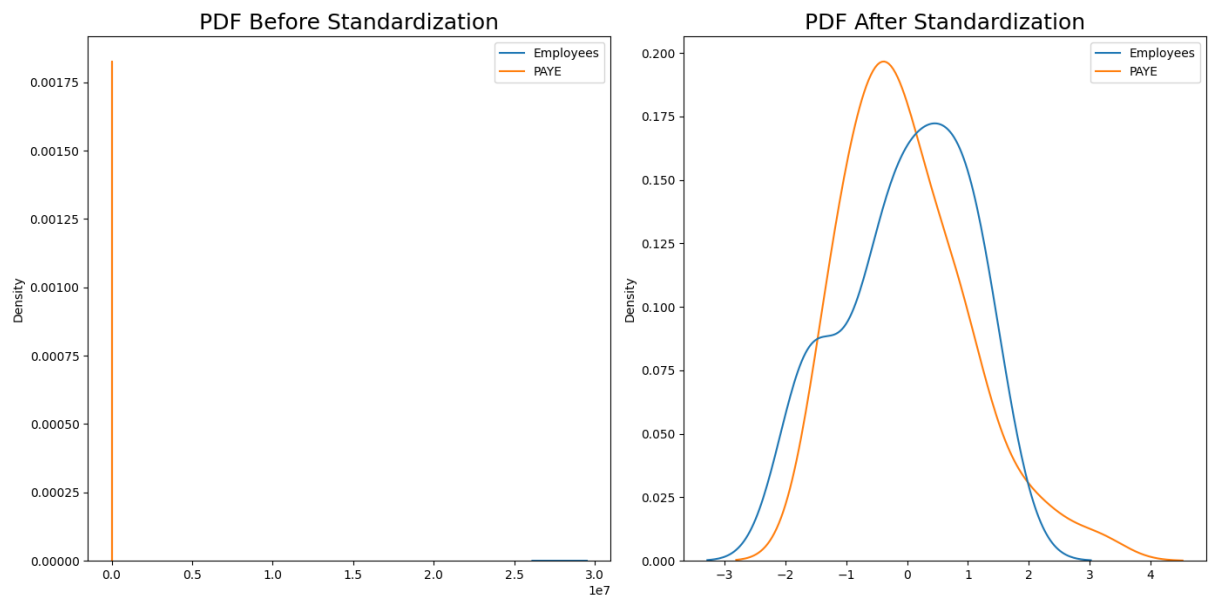


Figure 13: KDE distributions before and after StandardScaler, confirming mean = 0, std = 1.

TASK 4: Supervised ML algorithm

Summary

Two supervised machine learning models were used to predict the mean pay for June 2025: Linear Regression and Random Forest Regressor. The first model of linear regression assumes a linear relationship between the input and the target variable. It then finds coefficients that minimise prediction error. Its simplicity and ease of interpretation make it highly useful. Secondly, random forest is an ensemble learning method. It employs the construction of several decision trees from random subsets of data and features. The final predictions are made by averaging the results. Therefore, the model is able to find nonlinear relationships and interactions between variables.

This model choice allows a comparison between a straightforward linear approach and a more adaptable nonlinear model. Results from Task 2 already revealed significant variability in sector-level relationships, indicating a global linear assumption might be inadequate. Therefore, Random Forest is expected to perform better because it can capture more complex patterns across different areas.

The dataset was prepared through preprocessing for modelling. The Mean-Pay dataset was cleaned by encoding and removing unnecessary categorical columns. The sectors became individual observations. Additionally, both datasets were combined to match the input features and target variables. All observations before June 2025 were used for training, and June 2025 was used for testing. This method prevents data leakage and provides a realistic way to make predictions.

Several metrics were used to evaluate the model's performance: Mean Squared Error(MSE), Root Mean Squared Error(RMSE), Mean Absolute Error(MAE), and R^2 score. Those give a full picture of prediction accuracy and their explanatory power.

Linear regression outperforms the random forest model. It achieves lower errors and a significantly higher R^2 (0.9865), indicating the relationship is overall linear at this level of aggregation. Visual analysis backs this up even more, showing that Linear Regression predictions are closer to the perfect diagonal line. Random forest shows larger deviations in sectors with more complex wage dynamics.

The dataset characteristics help explain the performance difference. Although Task 2 showed variability across sectors. Linear Regression's strong performance indicates that the overall relationship between employment and pay remains largely linear at this level of aggregation. Consequently, the linearity assumption does not significantly impact the performance of the model. However, Random Forest may be less effective due to the relatively small dataset, which restricts its capacity to recognise intricate nonlinear patterns.

While both models perform well in stable sectors such as healthcare, they demonstrate greater inaccuracy in sectors such as manufacturing and retail, where external factors influence wages. This highlights a key limitation: the absence of macroeconomic variables (e.g. inflation and GDP), means it is difficult to make accurate predictions.

Linear Regression proves to be the more suitable model for this task, while Random Forest offers greater flexibility, its performance is limited in this case. This highlights that model performance depends on both algorithm choice and underlying data structure, rather than complexity alone.

Future improvements may involve integrating additional features, such as inflation or GDP indicators, alongside the utilisation of time-series-specific models (e.g., ARIMA or LSTM) to more effectively capture temporal dependencies.

Metric	Linear Regression	Random Forest
MSE	29870	1293366.25
RMSE	172.83	1137.26
MAE	151	828.25
R ²	0.9865	0.4151

Table 6: Metrics of supervised ML methods

Linear regression is significantly outperforming random forest, with nearly all variance being explained and it being suggested that the dataset lacks sufficient complexity for non-linear models ($R^2 = 0.9865$ vs. 0.4151). This calls into question the idea that more intricate models always perform better and emphasises the significance of matching model selection with the underlying data structure.

Visualisations supervised ML

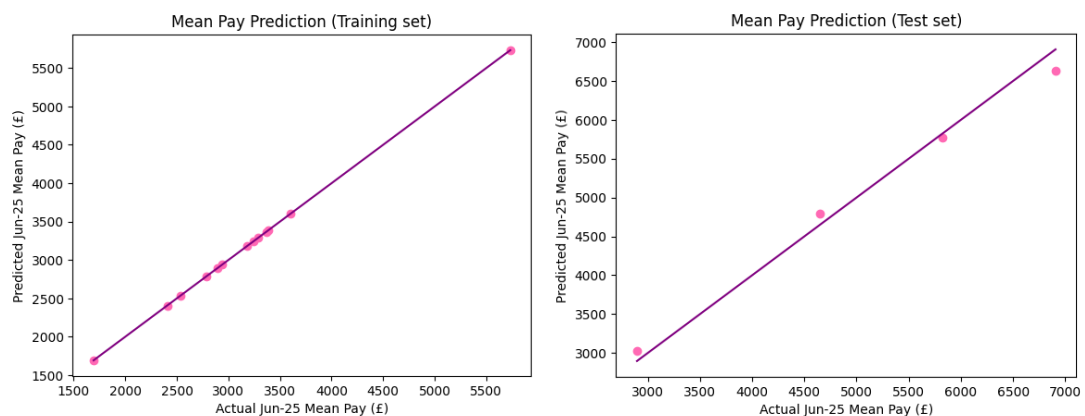


Figure 14: Linear Regression – training & test set

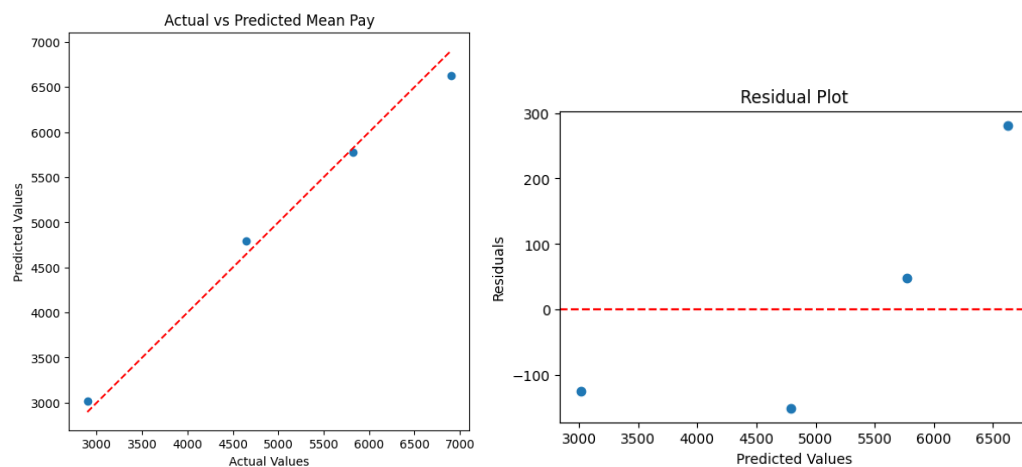


Figure 15: Linear regression – actual vs predicted mean pay & residual plot

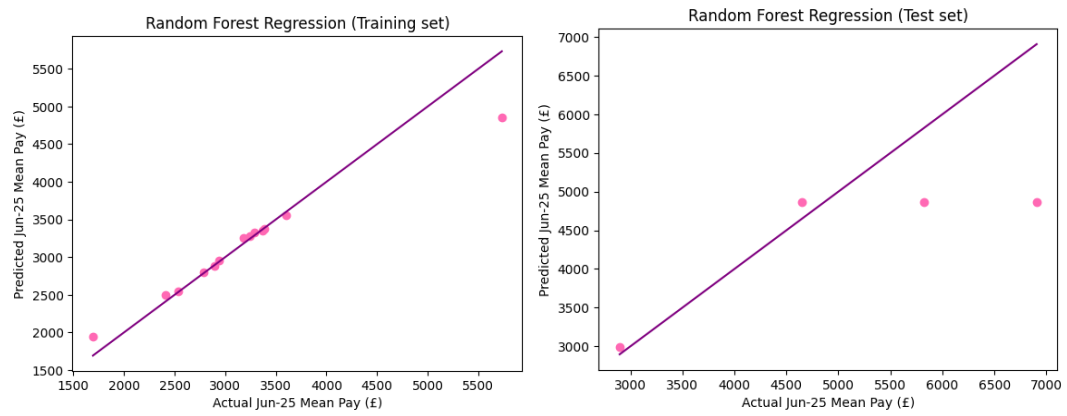


Figure 16: Random forest – training & test set

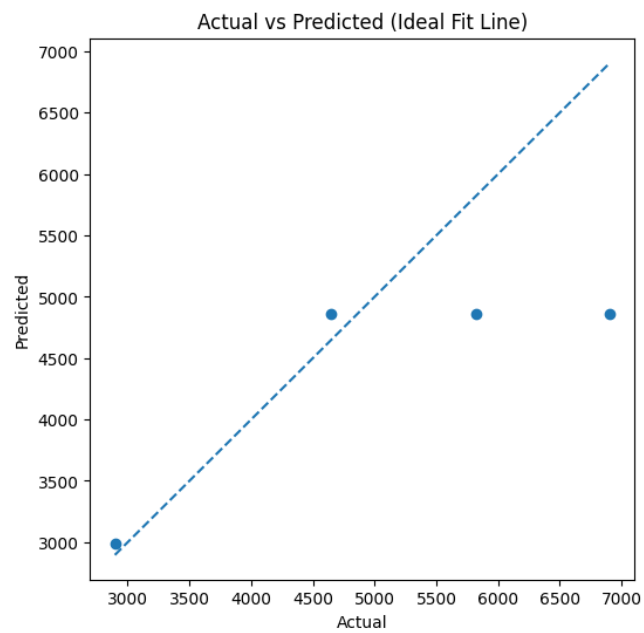


Figure 17: random forest – actual vs. predicted

TASK5: Unsupervised ML algorithm

Summary

K-Means clustering has been applied to the first 50 observations of the dataset to examine the relationship between Employees and PAYE. K-Means clustering was selected for its efficiency in grouping observations by minimising within-cluster variance. Distance-based clustering is appropriate because both variables are continuous. StandardScaler was used to standardise the features before clustering so that differences in scale did not affect the distance calculations.

Hierarchical methods offer a structured view of cluster relationships, and DBSCAN can identify non-spherical and noisy clusters. K-means was selected because of its computational efficiency, scalability and interpretability.

The optimal number of clusters was found by using both the elbow method and silhouette analysis. The elbow plot clearly shows an inflection point at $k = 4$, where the drop in the within-cluster sum of squares starts to level off. The silhouette score supports this, reaching its maximum (around 0.4434) at $k = 4$. This makes the selection stronger than methods relying on guessing.

The clusters reveal different ways employment and pay operate. The clusters show different employment-pay structures, from low-pay to high-pay sectors, which are based on differences in productivity and skill intensity. But the clusters have some overlap, especially in the middle range of values. This suggests that the relationship between Employees and PAYE is continuous rather than strictly segmented.

The moderate silhouette score suggests a limited separation and indicates that alternative methods, such as DBSCAN, may be more effective in capturing overlapping structures. These results are similar to those of Alexi et al. (2024), who showed that economic sub-populations often do not form distinct clusters.

The identified clusters are economically meaningful groupings. The sectors with high pay and high employment probably correspond to knowledge-intensive industries. The lower-pay clusters probably represent labour-intensive sectors. This shows how unsupervised learning can reveal economic structures that are not immediately visible through aggregate analysis.

Visualisations of unsupervised ML

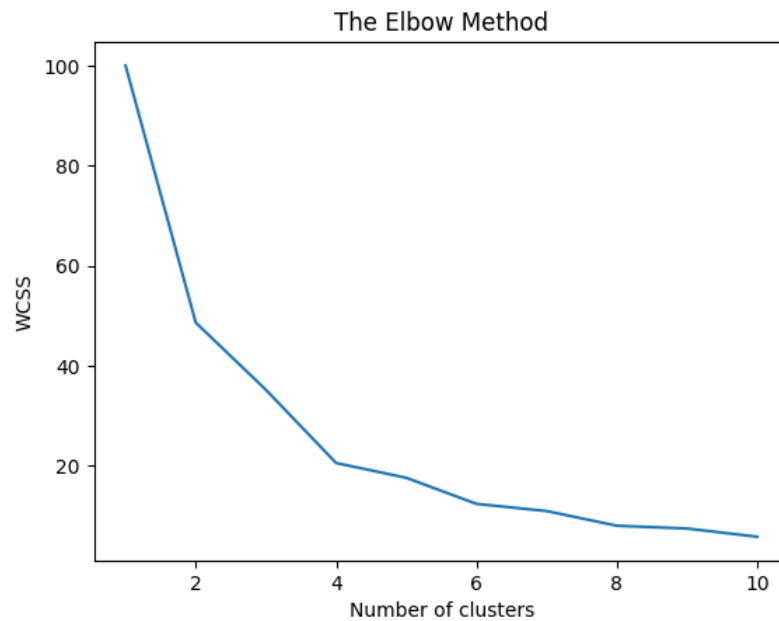


Figure 18: Elbow Method plot: WCSS against number of clusters ($k = 1$ to 10), used alongside the Silhouette Score to identify the optimal k .

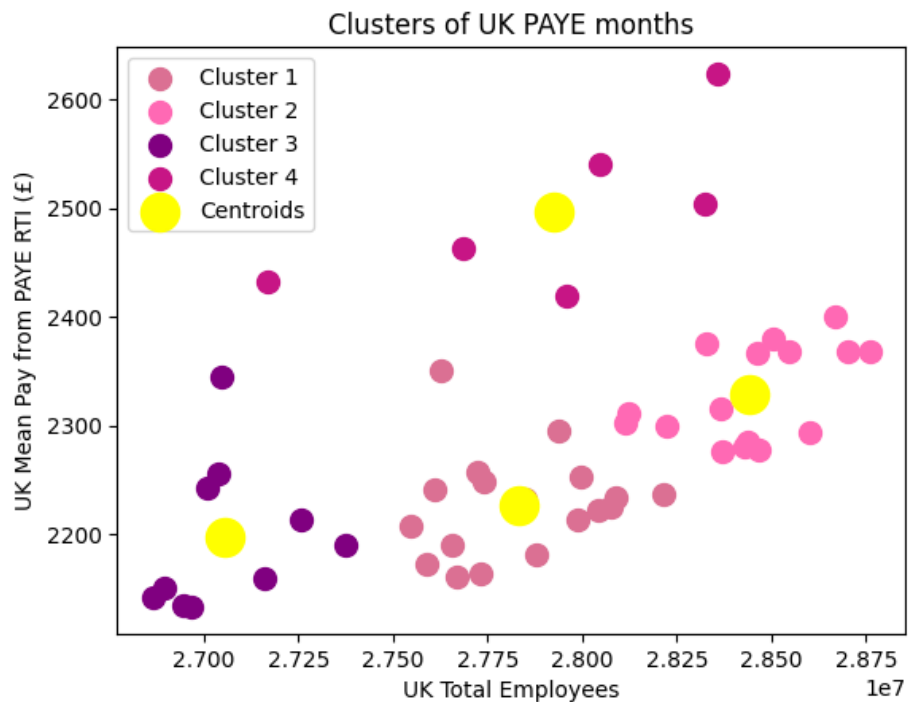


Figure 19: K-Means cluster scatter plot of UK total employees vs UK mean PAYE pay for the first 50 tuples, with cluster centroids marked in yellow.

TASK 6: Conclusion

This project demonstrated a full-cycle machine learning pipeline, combining exploratory data analysis with predictive modelling to examine the relationship between employment and PAYE across sectors.

The analysis revealed strong sectoral patterns that were accurately captured during the modelling-stage. Linear regression achieved the highest predictive performance and achieved a significantly higher R^2 score than Random Forest. This indicates that the relationship between employment and PAYE is largely linear at this level of aggregation.

The clustering analysis was used to highlight the underlying group structures present in the data. Using the Elbow Method and Silhouette Score to select four clusters strengthened the reliability of the results. However, the partial overlap between the clusters indicates the relationship is continuous rather than strictly segmented.

In the future, improvements could include incorporating additional features, such as macroeconomic indicators(i.e.,inflation, GDP), and applying time-series models to better capture seasonal relations. During implementation, minor challenges were encountered, such as data formatting and alignment(i.e.,merging datasets and preparing inputs). These were successfully resolved through careful pre-processing and validation. This demonstrates the importance of data preparation in ML workflows.

The fourth task provided valuable insights, highlighting that simpler models can outperform more complex ones when the data structure is primarily linear. Task 5 was also intriguing, as K-means clustering revealed hidden temporal structure in the unlabelled data. This shows how unsupervised methods can generate hypotheses for further analysis. Overall, the project shows the importance of combining data analysis with ML to gain actionable insights and make accurate predictions.

A key limitation of this study is its reliance on a simplified dataset containing induced errors, which may not accurately reflect real-world PAYE dynamics. Furthermore, the models assume relatively stable relationships over time, but these may be affected by economic shocks. Future research could build on this work by incorporating macroeconomic indicators and applying advanced time series models, such as long short-term memory(LSTM) or autoregressive integrated moving average(ARIMA), to better capture temporal dependencies and structural changes.

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